Quiz 2 Solutions, MA2031, Oct 9, 2023

1. Let $A = \begin{bmatrix} -3 & -2 & 0 & -1 \\ 8 & 5 & 1 & 2 \\ 7 & 4 & 2 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}$. Construct bases for

(a) $R(A)$ and (b) $N(A)$. [5]

Solution: $RREF(A) = \begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$. [2]

Basis for $R(A)$ is $\left\{ \begin{bmatrix} -3 \\ 8 \\ 7 \\ -2 \end{bmatrix}, \begin{bmatrix} -2 \\ 5 \\ 4 \\ -1 \end{bmatrix} \right\}$. [1]

Basis for $N(A)$ is $\left\{ \begin{bmatrix} 2 \\ -3 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 2 \\ 0 \\ -1 \end{bmatrix} \right\}$. [2]

2. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 1 & 2 \end{bmatrix}$. Consider the ordered bases

$C = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\}$ and $E = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\}$ for $\mathbb{R}^{3\times 1}$.

Compute the coordinate matrix $[A]_{E,C}$. [5]

(Note: $[A]_{E,C}$ is the matrix that satisfies $[Ax]_E = [A]_{E,C}[x]_C$ for all $x \in \mathbb{R}^{3\times 1}$.)

Solution: We compute the RREF of the augmented matrix with its columns as $w_1, w_2, w_3, Av_1, Av_2, Av_3$, where v_1, v_2, v_3 are vectors in C and w_1, w_2, w_3 are vectors in E. Now,

$Av_1 = \begin{bmatrix} 2 \\ 4 \\ 5 \end{bmatrix}$, $Av_2 = \begin{bmatrix} 3 \\ 6 \\ 6 \end{bmatrix}$, $Av_3 = \begin{bmatrix} 2 \\ 4 \\ 4 \end{bmatrix}$. [1]

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 2 & 3 & 2 \\ -1 & 0 & 1 & 4 & 6 & 4 \\ 0 & 1 & -1 & 5 & 6 & 4 \end{array}\right] \xrightarrow{RREF} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & 9 & 6 \\ 0 & 1 & 0 & 16 & 21 & 14 \\ 0 & 0 & 1 & 11 & 15 & 10 \end{array}\right].$$ [3]

So, $[A]_{E,C} = \begin{bmatrix} 7 & 9 & 6 \\ 16 & 21 & 14 \\ 11 & 15 & 10 \end{bmatrix}$. [1]

①

$$\begin{bmatrix} -3 & -2 & 0 & -1 \\ 8 & 5 & 1 & 2 \\ 7 & 4 & 2 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 2/3 & 0 & 1/3 \\ 8 & 5 & 1 & 2 \\ 7 & 4 & 2 & 1 \\ -2 & -1 & -1 & 0 \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} 1 & 0 & 2 & -1 \\ 0 & -1/3 & 1 & -2/3 \\ 0 & -2/3 & 2 & -4/3 \\ 0 & 1/3 & -1 & 2/3 \end{bmatrix} \Leftarrow \begin{bmatrix} 1 & 2/3 & 0 & 1/3 \\ 0 & -1/3 & 1 & -2/3 \\ 0 & -2/3 & 2 & -4/3 \\ 0 & 1/3 & -1 & 2/3 \end{bmatrix}$$

$$\Downarrow$$

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & -2 & 6 & -4 \\ 0 & -1 & -3 & 2 \end{bmatrix} \Rightarrow \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & -3 & 2 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

R(A) N(A)

2. Let $A = \begin{bmatrix} 1 & 1 & 1 \\ 2 & 2 & 2 \\ 3 & 1 & 2 \end{bmatrix}$. Consider the ordered bases

$$C = \left\{ \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} \right\} \text{ and } E = \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right\} \text{ for } \mathbb{R}^{3\times 1}.$$

Compute the coordinate matrix $[A]_{E,C}$. [5]

(Note: $[A]_{E,C}$ is the matrix that satisfies $[Ax]_E = [A]_{E,C}[x]_C$ for all $x \in \mathbb{R}^{3\times 1}$.)

$$P = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix} \quad Q = \begin{bmatrix} 1 & -1 & 1 \\ -1 & 0 & 1 \\ 0 & 1 & -1 \end{bmatrix}.$$

$$[A]_{E,C} = Q^T A P \Rightarrow \left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 2 & 3 & 2 \\ -1 & 0 & 1 & 4 & 6 & 4 \\ 0 & 1 & -1 & 5 & 6 & 4 \end{array}\right]$$

$\downarrow E_1[2,1]$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 2 & 3 & 2 \\ 0 & 1 & 2 & 6 & 9 & 6 \\ 0 & 1 & -1 & 5 & 6 & 4 \end{array}\right]$$

$\downarrow E_1[2]$

$$\left[\begin{array}{ccc|ccc} 1 & -1 & 1 & 2 & 3 & 2 \\ 0 & 1 & -2 & -6 & -9 & -6 \\ 0 & 1 & -1 & 5 & 6 & 4 \end{array}\right]$$

$\downarrow E_1[1,2]$
$\downarrow E_1[3,2]$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & -1 & -4 & -6 & -4 \\ 0 & 1 & -2 & -6 & -9 & -6 \\ 0 & 0 & 1 & 11 & 15 & 10 \end{array}\right]$$

$\downarrow E_1[1,3]$
$\downarrow E_2[2,3]$

$$\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 7 & 9 & 6 \\ 0 & 1 & 0 & 16 & 21 & 14 \\ 0 & 0 & 1 & 11 & 15 & 10 \end{array}\right]$$

Quiz 2 Solutions, MA2031, Oct 9, 2023

3. Compute the QR-factorization of the matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}. \qquad [6]$$

Solution: Gram-Schmidt orthonormalization of the columns of A yield the vectors

$$\begin{bmatrix} 1/\sqrt{3} \\ 1/\sqrt{3} \\ 0 \\ 1/\sqrt{3} \end{bmatrix}, \quad \begin{bmatrix} -1/\sqrt{15} \\ 2/\sqrt{15} \\ 3/\sqrt{15} \\ -1/\sqrt{15} \end{bmatrix}, \quad \begin{bmatrix} 1/\sqrt{3} \\ -1/\sqrt{3} \\ 1/\sqrt{3} \\ 0 \end{bmatrix}. \qquad [3]$$

Then, $Q = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{15} & 1/\sqrt{3} \\ 1/\sqrt{3} & 2/\sqrt{15} & -1/\sqrt{3} \\ 0 & 3/\sqrt{15} & 1/\sqrt{3} \\ 1/\sqrt{3} & -1/\sqrt{15} & 0 \end{bmatrix}$. [1]

Then, $R = Q^*A = \dfrac{1}{\sqrt{3}} \begin{bmatrix} 3 & 1 & 2 \\ 0 & \sqrt{5} & -\sqrt{5} \\ 0 & 0 & 3 \end{bmatrix}$. [2]

4. Determine all least squares solutions of the system of linear equations:

$x + z = 6, \quad x + y - z = 0, \quad x + 2y - 3z = 0.$ [4]

Solution: The system is $AX = b$, where $A = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & -1 \\ 1 & 2 & -3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$, $b = \begin{bmatrix} 6 \\ 0 \\ 0 \end{bmatrix}$.

We then have

$$A^*A = \begin{bmatrix} 3 & 3 & -3 \\ 3 & 5 & -7 \\ -3 & -7 & 11 \end{bmatrix}, \quad A^*b = \begin{bmatrix} 6 \\ 0 \\ 6 \end{bmatrix}. \qquad [1]$$

$$[A^*A|A^*b] \xrightarrow{RREF} \left[\begin{array}{ccc|c} 1 & 0 & 1 & 5 \\ 0 & 1 & -2 & -3 \\ 0 & 0 & 0 & 0 \end{array}\right]. \qquad [2]$$

Thus, the least squares solutions are given by

$$\begin{bmatrix} 5 \\ -3 \\ 0 \end{bmatrix} + \alpha \begin{bmatrix} 1 \\ -2 \\ -1 \end{bmatrix} \quad \text{for } \alpha \in \mathbb{R}. \qquad [1]$$

3. Compute the QR-factorization of the matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}. \quad [6]$$

$u_1 = [1\ 1\ 0\ 1]^t \quad u_2 = [0\ 1\ 1\ 0]^t \quad u_3 = [2\ -1\ 0\ 1]^t$

$v_1 = u_1 = [1\ 1\ 0\ 1]^t \quad w_1 = [1/\sqrt{3}\ \ 1/\sqrt{3}\ \ 0\ \ 1/\sqrt{3}]^t$

$v_2 = u_2 - \langle u_2 \cdot w_1 \rangle w_1 = [0\ 1\ 1\ 0]^t - 1/\sqrt{3}\,[1/\sqrt{3}\ \ 1/\sqrt{3}\ \ 0\ \ 1/\sqrt{3}]^t$

$= [-1/3\ \ 2/3\ \ 1\ \ -1/3]^t$

$w_2 = [-1/\sqrt{15}\ \ 2/\sqrt{15}\ \ 3/\sqrt{15}\ \ -1/\sqrt{15}]^t$

$v_3 = u_3 - \langle u_3 \cdot w_1 \rangle w_1 - \langle u_3 \cdot w_2 \rangle w_2$

$= [2\ -1\ 0\ 1]^t - 2/\sqrt{3}\,[1/\sqrt{3}\ \ 1/\sqrt{3}\ \ 0\ \ 1/\sqrt{3}]^t + 5/\sqrt{15}\,[-1/\sqrt{15}\ \ 2/\sqrt{15}\ \ 3/\sqrt{15}\ \ -1/\sqrt{15}]^t$

$= [1\ -1\ 1\ 0]^t \quad w_3 = [1/\sqrt{3}\ \ -1/\sqrt{3}\ \ 1/\sqrt{3}\ \ 0]^t$

$$Q = \begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{15} & 1/\sqrt{3} \\ 1/\sqrt{3} & 2/\sqrt{15} & -1/\sqrt{3} \\ 0 & 3/\sqrt{15} & 1/\sqrt{3} \\ 1/\sqrt{3} & -1/\sqrt{15} & 0 \end{bmatrix}$$

$$R = Q^T A = \begin{bmatrix} 1/\sqrt{3} & 1/\sqrt{3} & 0 & 1/\sqrt{3} \\ -1/\sqrt{15} & 2/\sqrt{15} & 3/\sqrt{15} & -1/\sqrt{15} \\ \cdots & \cdots & \cdots & \cdots \end{bmatrix} \begin{bmatrix} 1 & 0 & 2 \\ 1 & 1 & -1 \\ 0 & 1 & 0 \\ \cdots & \cdots & \cdots \end{bmatrix}$$

$$\begin{bmatrix} 1/\sqrt{3} & -1/\sqrt{3} & 1/\sqrt{3} \\ 0 & 1/\sqrt{5} & -1/\sqrt{5} \\ 1 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} \sqrt{3} & 1/\sqrt{3} & 2/\sqrt{3} \\ 0 & 5/\sqrt{5} & -5/\sqrt{5} \\ 0 & 0 & \sqrt{3} \end{bmatrix}$$

Indian Institute of Technology Madras
Department of Mathematics
MA2031-Linear Algebra for Engineers
Quiz-I

Time : 8.00 AM-8.50 AM **Date : Feb 20, 2023** **Max Marks : 20**

Instructions:

(i) Answer **ALL** questions.

(ii) Write your name and **Roll number** on the answer book.

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1. Determine if the following sets are subspaces of $\mathbb{R}^3$. Justify your answers.

   (a) $W = \{(x, y, z) \mid x^3 = y^3\}$. [3]

   (b) $W = \{(x, y, z) \mid x^2 = y^2 + z^2\}$. [2]

2. Let $V = span\{(1, 0, 0), (1, -1, 0)\}$ and $W = span\{(0, 1, 0), (0, -1, 1)\}$.

   (a) Find the dimensions of vector spaces $V + W$ and $V \cap W$. [3]

   (b) Identify the vector space $V + W$. [1]

   (c) Is $V + W$ a direct sum?. [1]

3. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$. Is the solution set W of the linear system $Ax = 0$ a subspace of $\mathbb{R}^3$? Justify your answer. If yes, find a basis for W. [5]

4. Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be defined by $T(x, y) = (x - y, 0, x + y)$. Let

$$B_1 = \{(1, 2), (1, 1)\} \text{ and } B_2 = \{(1, 1, 1), (1, 1, 0), (0, 1, 1)\}$$

   be ordered bases for $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively. Find the matrix representation $M_{B_1 B_2}(T)$ of the linear map T with respect to the bases B_1 and B_2. [5]

2. Let $V = span\{(1,0,0),(1,-1,0)\}$ and $W = span\{(0,1,0),(0,-1,1)\}$.

(a) Find the dimensions of vector spaces $V + W$ and $V \cap W$.

(b) Identify the vector space $V + W$.

(c) Is $V + W$ a direct sum?.

$V+W \supseteq span\{(1,0,0),(0,1,0),(0,-1,1)\}$

$\supseteq \mathbb{R}^3$

But $V+W \subseteq \mathbb{R}^3 \rightarrow V+W = \mathbb{R}^3$ (dimension = 3)

$\dim(V+W) = \dim(V) + \dim(W) - \dim(V \cap W)$

$\Rightarrow \dim(V \cap W) = 1$

$\Rightarrow V \cap W \neq 0$

$\Rightarrow V+W$ isn't a direct sum

3. Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 3 & 6 & 9 \end{bmatrix}$. Is the solution set W of the linear system $Ax = 0$ a subspace of $\mathbb{R}^3$? Justify your answer. If yes, find a basis for W. [5]

If x_1, x_2 are solutions:

$Ax_1 = 0$, $Ax_2 = 0$

$A(x_1 + \alpha x_2) = Ax_1 + \alpha Ax_2 = 0 \Rightarrow W$ is subspace of $\mathbb{R}^3$

$\boxed{x+2y+3z=0}$

$(-2\alpha - 3\beta, \alpha, \beta) \rightarrow$ solution

$= \alpha(-2,1,0) + \beta(-3,0,1)$

Basis: $(-2,1,0)$ $(-3,0,1)$

4. Let $T : \mathbb{R}^2 \to \mathbb{R}^3$ be defined by $T(x, y) = (x - y, 0, x + y)$. Let

$$B_1 = \{(1, 2), (1, 1)\} \text{ and } B_2 = \{(1, 1, 1), (1, 1, 0), (0, 1, 1)\}$$

be ordered bases for $\mathbb{R}^2$ and $\mathbb{R}^3$ respectively. Find the matrix representation $M_{B_1 B_2}(T)$ of the linear map T with respect to the bases B_1 and B_2. [5]

$T[u_1] = a[1,1,1] + b[1,1,0] + c[0,1,1]$

$\Rightarrow (-1,0,3) = (a+b, a+b+c, a+c)$

$\Rightarrow (a,b,c) = [2,-3,1]$

$(0,0,2) = (p+q, p+q+r, p+r)$

$\Rightarrow (p,q,r) = [2,-2,0]$

$$M_{B_1 B_2}[T] = \begin{bmatrix} 2 & 2 \\ -3 & -2 \\ 1 & 0 \end{bmatrix}$$

Department of Mathematics, IIT Madras
MA 2031 - Linear Algebra for Engineers
Quiz I
Date: October 12, 2020. Time: 50 minutes Max. marks: 20

Answer all questions. Each question carries 4 marks.

1. Which of the following subsets of $\mathbb{R}^3$ are subspaces? Justify your conclusions.
   (a) $W = \{(x, y, z) : x^3 = y^3\}$.
   (b) $W = \{(x, y, z) : x^4 = y^4\}$.
2. Let A and B be subsets of a vector space V. Let "sp" denote the linear span. Show that
   (a) $\text{sp}(\text{A} \cap \text{B}) \subseteq \text{sp}(\text{A}) \cap \text{sp}(\text{B})$.
   (b) Is the inclusion $\text{sp}(\text{A}) \cap \text{sp}(\text{B}) \subseteq \text{sp}(\text{A} \cap \text{B})$ true? Justify your answer.
3. Determine the relationship between a, b if the following vectors in $\mathbb{R}^4$ are linearly *independent*.
$$\{(1, 1, 1, a), (1, 1, 0, 1), (-1, 2, b, -2), (1, 0, 0, 1)\}.$$
4. Suppose that subspaces U, W of $\mathbb{R}^3$ are defined by
$$U = \{(x, y, z) : x + y + 2z = 0\}, \; W = \{(x, y, z) : x - y - 2z = 0\}.$$
Find bases for $U \cap W$ and $U + W$. Justify your answers.
5. Using Gram-Schmidt process, determine *all the vectors* that are orthogonal to the vector $(0, 1, 1)$, relative to the standard inner product in $\mathbb{R}^3$.

5. Using Gram-Schmidt process, determine *all the vectors* that are orthogonal to the vector $(0, 1, 1)$, relative to the standard inner product in $\mathbb{R}^3$.

Say (x, y, z)

$x \cdot 0 + y \cdot 1 + z \cdot 1 = 0$

$\Rightarrow y + z = 0$

$\Rightarrow (x, y, -y)$

$(2, 1, -1)$
$(3, 1, -1)$

take 2 vectors: $(1, 0, 0)$ $(0, 1, -1)$

$v_1 = (1, 0, 0)$

$v_2 = (0, 1, -1) - \frac{\langle 0 \rangle}{1} (1, 0, 0) = (0, 1, -1)$

$\Rightarrow$ They're already O.N

span $(y + z = 0) \rightarrow$ subspace

Department of Mathematics, IIT Madras
MA 2031 - Linear Algebra for Engineers
Quiz II

Answer all questions. Time: 50 minutes Max. marks: 20

1. (a) Find a least squares solution to the following system of equations:

$$3x + y = 1,\ x + 2y = 0,\ 2x - y = -2.$$

(3 marks)

(b) Let $A \in \mathbb{C}^{m\times n}$ and $b \in \mathbb{C}^m$. When does the (possibly inconsistent) system $Ax = b$ have a *unique* least squares solution? Justify your answer.

(2 marks)

2. Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ be defined by

$$T(a, b, c) = (a - b, b - c, a - c, a - 2b + c),\ a, b, c \in \mathbb{R}.$$

Find bases for the range space and the null space of T.

(4 marks)

3. Let $T : V \to V$ be a linear map satisfying $T^2 = T$. Show that
(*a*) $V = R(T) + N(T)$
(*b*) $R(T) \cap N(T) = \{0\}$.

(2 + 2 marks)

4. Let $T : \mathbb{F}^n \to \mathbb{F}^n$ be defined by $T(z_1, z_2, \ldots, z_n) = (0, z_1, \ldots, z_{n-1})$. Obtain the formula for $T^*(w),\ w \in \mathbb{F}^n$.

(3 marks)

5. Let V denote the vector space of all polynomials in the single variable t, with coefficients from $\mathbb{R}$. Suppose that P_n and P_{n+1} are the subspaces of V consisting of polynomials of degree at most n and $n + 1$, respectively.

(a) Let $T : P_n \to P_{n+1}$ be the linear map defined by $T(p(t)) = \int_0^t p(x)\,dx$. Compute the nullity and the rank of T.

(b) Let $S : V \to V$ be the linear map defined by $S(p(t)) = \int_0^t p(x)\,dx$. Is S an isomorphism? Justify your conclusion.

(2 + 2 marks)

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2. Let $T : \mathbb{R}^3 \longrightarrow \mathbb{R}^4$ be defined by

$$T(a, b, c) = (a - b, b - c, a - c, a - 2b + c), \ a, b, c \in \mathbb{R}.$$

Find bases for the range space and the null space of T. (4 marks)

$T(a,b,c) = a(1,0,1,1) + b(-1,1,0,-2) + c(0,-1,-1,1)$

$$T = \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 0 & -1 \\ 1 & -2 & 1 \end{bmatrix} \qquad \begin{bmatrix} 1 & -1 & 0 \\ 0 & 1 & -1 \\ 0 & 1 & -1 \\ 0 & -1 & 1 \end{bmatrix}$$

$R: [1,0,1,1]^t,$

$[-1,1,0,-2]^t$

$$\leftarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$N: [-1,-1,0,-1]^t$

1. (a) Find a least squares solution to the following system of equations:

$$3x + y = 1, \ x + 2y = 0, \ 2x - y = -2.$$

$$\begin{bmatrix} 3 & 1 \\ 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix}$$

$$A^*A = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 3 & 1 \\ 1 & 2 \\ 2 & -1 \end{bmatrix} \begin{bmatrix} 14 & 3 \\ 3 & 14 \end{bmatrix}$$

$$A^*b = \begin{bmatrix} 3 & 1 & 2 \\ 1 & 2 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ -2 \end{bmatrix} = \begin{bmatrix} -1 \\ 3 \end{bmatrix}$$

$$\left[\begin{array}{cc|c} 14 & 3 & -1 \\ 3 & 14 & 3 \end{array}\right] = \begin{bmatrix} 1 & 3/14 & -1/14 \\ 3 & 14 & 3 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 3/14 & -1/14 \\ 0 & 187/14 & 45/14 \end{bmatrix} \rightarrow \left[\begin{array}{cc|c} 1 & 0 & -23/187 \\ 0 & 1 & 45/187 \end{array}\right]$$

↓ Solutions